1 Early Warning System

1.1 Motivation: From Vulnerability Measurement to Anticipation

The risk and vulnerability metrics developed in the preceding sections provide, at each forecast origin, a comprehensive snapshot of the conditional distribution of future Brent returns. The Kullback–Leibler divergence, in particular, measures how far the conditional density has departed from its reference state, quantifying the degree to which the market's return distribution has moved away from the well-behaved regime in which traditional risk metrics remain reliable. A natural and economically consequential question follows: does this departure carry predictive content? If a spike in tail entropy today foreshadows large oil price movements tomorrow, then the entropy measure ceases to be a contemporaneous diagnostic and becomes an early warning indicator, one that can alert risk managers and policymakers to emerging vulnerabilities before they materialise in realised returns.

The distinction between a contemporaneous risk measure and a leading indicator is fundamental in practice. A risk manager at a commodity trading desk who observes elevated left-tail entropy at the close of trading today faces two very different operational realities depending on whether that signal is coincident or anticipatory. If the entropy spike merely reflects the same information already priced into today's returns, it adds nothing to the decision-making process that was not already available from the return series itself. If, however, the entropy spike systematically precedes large return movements by several trading days, it provides an actionable window during which hedging positions can be adjusted, margin requirements can be recalibrated, and contingency plans can be activated before the stress event arrives. The difference between these two scenarios is the difference between a thermometer and a barometer: both measure atmospheric conditions, but only the barometer tells you what is coming.

This section formalises the early warning hypothesis and subjects it to two independent statistical tests. The analysis proceeds in three stages. First, the choice of the entropy indicator and its configuration are motivated on economic and statistical grounds. Second, the Cross-Correlation Function is deployed to identify the lead-lag structure between the entropy measures and Brent returns, establishing whether the peak correlation occurs at a positive lag, which would confirm that the entropy series leads the target. Third, the Granger causality test is applied to determine whether lagged values of the entropy series carry statistically significant predictive information for future returns beyond the information already contained in the returns' own history.

1.2 Choice of the Early Warning Indicator

The entropy measures constructed in the preceding chapter offer several candidate indicators for the early warning system. Two design decisions must be made: the choice of the reference baseline against which the conditional density is compared, and the choice of the estimation window used to produce the entropy series.

Regarding the baseline, the framework provides two alternatives. The first is the unconditional Johnson SU density fitted by Maximum Likelihood to the historical returns, which is the reference used in the original Vulnerable Growth framework of Adrian, Boyarchenko, and Giannone (2019). The second is a Normal density calibrated to the historical mean and standard deviation of the return series. The unconditional JSU baseline measures genuine regime shifts in the conditional distribution, because the reference itself already accommodates the skewness and heavy tails that characterise oil returns. The Normal baseline, by contrast, measures the departure from Gaussianity, a broader concept that captures not only regime shifts but also the structural non-normality that is a permanent feature of commodity return distributions.

For the purpose of early warning, the Normal baseline is the more informative choice. The empirical observation motivating this decision is that the historical distribution of Brent returns is approximately Gaussian in its central mass, a property that is well documented in the commodity finance literature and that the Kolmogorov–Smirnov tests in the distribution fitting chapter confirm. The tails, however, deviate substantially from normality during episodes of market stress. The KL divergence against a Normal reference therefore functions as a detector of tail fattening: when the conditional density closely resembles a Gaussian, the divergence is near zero; when the tails fatten in response to geopolitical escalation, speculative volatility, or demand-side shocks, the divergence spikes. This spike-and-revert dynamic is precisely the pattern that an early warning system should detect.

The second design decision concerns the estimation window. The expanding window accumulates all historical data up to each forecast origin, which produces a smooth entropy series but dilutes the influence of recent observations as the sample grows. The rolling window, by contrast, uses only the most recent $W = 1{,}008$ observations (four trading years), producing a more responsive signal that adapts quickly to regime changes. For early warning purposes, the rolling window is preferable on three grounds. First, it is computationally more efficient, with cost per iteration of order $O(W)$ rather than $O(t)$, which matters when the system is deployed in a real-time monitoring context. Second, it better captures regime transitions, because the influence of a departing regime decays as its observations leave the window rather than being permanently embedded in the expanding average. Third, the rolling window is the standard convention in regulatory risk management, as codified in the Basel III internal-model framework (Basel Committee on Banking Supervision, 2019), and adopting it here ensures consistency with the VaR and Expected Shortfall computations that use the same window size.

The forecast horizon for the entropy computation is set at $h = 5$ trading days, corresponding to one trading week. This is a deliberate choice that balances two competing considerations. At shorter horizons such as $h = 1$ or $h = 2$, the quantile regression model exhibits the strongest predictive power, as demonstrated by the pinball loss comparisons in the direct forecasting chapter. However, a one-day-ahead entropy signal provides too narrow a window for operational response: by the time the signal is observed at the close of trading, the event it anticipates may already be underway the following morning. At longer horizons such as $h = 22$ (one trading month), the quantile regression coefficients become noisier and the MDE fitting inherits that noise, degrading the quality of the conditional density and, consequently, the reliability of the entropy measure. The five-day horizon occupies a middle ground where the conditional density retains meaningful predictive content while providing a sufficiently wide anticipation window for risk management action. Moreover, the specification diagnostics conducted in the model selection chapter confirm that the standard quantile regression, without CAViaR augmentation, performs best at this intermediate horizon, consistent with the observation that CAViaR's autoregressive persistence is most valuable at the shortest horizons where serial dependence in the quantile process is strongest.

The early warning indicator is therefore defined as the rolling-window KL divergence of the conditional Johnson SU density against the Normal baseline at $h = 5$, decomposed into three components: the full-distribution entropy, the left-tail entropy (capturing downside vulnerability), and the right-tail entropy (capturing upside vulnerability). The decomposition follows the convention established in the risk metrics chapter, with the tail mark set at $\tau = 0.95$ so that the left-tail component captures the divergence below the 5th percentile and the right-tail component captures the divergence above the 95th percentile of the Normal reference.

1.3 The Cross-Correlation Function

The first diagnostic tool deployed to assess the anticipatory content of the entropy measures is the Cross-Correlation Function (CCF). The CCF is a standard instrument in the composite-indicator literature for identifying lead-lag relationships between economic time series, and its application to the early warning problem is direct: if the entropy series leads Brent returns, the CCF will exhibit its peak at a positive lag, indicating that today's entropy is most strongly correlated with future returns rather than with contemporaneous or past returns.

The CCF between two stationary time series $\{X_t\}$ and $\{Y_t\}$ at lag h is defined as the normalised cross-covariance:

$$r_{XY}(h) = \frac{c_{XY}(h)}{\sqrt{c_{XX}(0) \cdot c_{YY}(0)}}, \tag{1}$$

where $c_{XY}(h)$ is the sample cross-covariance at lag h. For non-negative lags ($h \geq 0$), the cross-covariance is computed as

$$c_{XY}(h) = \frac{1}{N} \sum_{t=1}^{N-h} (X_t - \bar{X})(Y_{t+h} - \bar{Y}), \tag{2}$$

and for negative lags ($h < 0$) the indices are reversed:

$$c_{XY}(h) = \frac{1}{N} \sum_{t=1-h}^{N} (X_t - \bar{X})(Y_{t+h} - \bar{Y}), \tag{3}$$

where $\bar{X}$ and $\bar{Y}$ are the sample means, and N is the number of observations (Brockwell and Davis, 1991, Chapter 11). The denominator in Equation (1) normalises the cross-covariance by the product of the standard deviations $\sqrt{c_{XX}(0)}$ and $\sqrt{c_{YY}(0)}$, ensuring that the CCF is bounded in $[-1, 1]$ and directly interpretable as a correlation coefficient.

The sign convention adopted throughout is that $h > 0$ means the indicator X leads the target Y by h periods. This convention is standard in the composite-indicator literature (Bujosa, García-Ferrer, and de Juan, 2013) and has a clear economic interpretation: if the CCF between left-tail entropy and Brent returns peaks at $h^* = +16$, then today's left-tail entropy is most strongly correlated with Brent returns 16 trading days hence.

Statistical significance of the CCF at each lag is assessed using Bartlett's approximation. Under the null hypothesis that the two series are independent, the sampling distribution of $r_{XY}(h)$ is approximately Normal with mean zero and standard deviation $1/\sqrt{N}$ for large N (Bartlett, 1946). The 95 percent confidence band is therefore $\pm 1.96/\sqrt{N}$, and any CCF value exceeding this band in absolute value is considered statistically significant at the 5 percent level.

Three interpretive cases arise from the position of the peak correlation h^*:

Case 1: Leading indicator ($h^* > 0$). The peak correlation occurs when the entropy series is aligned with future returns. This is the desired outcome for an early warning system, because it implies that the entropy signal today is most informative about returns h^* periods ahead. The magnitude of h^* determines the anticipation horizon: the number of trading days of advance notice that the indicator provides.

Case 2: Coincident indicator ($h^* = 0$). The peak correlation occurs at zero lag, meaning the entropy series moves in lockstep with contemporaneous returns. While mathematically valid, a coincident indicator has no predictive value: it tells the risk manager what is happening today, not what will happen tomorrow.

Case 3: Lagging indicator ($h^* < 0$). The peak correlation occurs when the entropy series is aligned with past returns, indicating that the indicator reacts to price movements after they have occurred. A lagging indicator cannot serve as an early warning system.

Both the entropy indicators and the Brent return target are first-differenced before the CCF is computed, to ensure stationarity. Although the return series is stationary by construction (it is a first difference of log prices), the entropy series can exhibit persistent level shifts during prolonged episodes of market stress, and differencing removes these non-stationary components while preserving the lead-lag dynamics of interest.

1.4 Granger Causality

The CCF identifies the lag at which the correlation between two series is maximised, but correlation does not establish predictive content in a rigorous econometric sense. Two series can be highly correlated at a positive lag because they are both driven by a common third factor, without the leading series carrying any incremental predictive information beyond what the target's own history already provides. The Granger causality test (Granger, 1969) addresses this limitation by asking a sharper question: do lagged values of the entropy indicator improve the forecast of Brent returns beyond the forecast that can be constructed from the returns' own lagged values alone?

The test is implemented by comparing two nested linear models. The restricted model regresses the target series on its own lagged values:

$$Y_t = \alpha_0 + \sum_{j=1}^{p} \alpha_j Y_{t-j} + e_t, \tag{4}$$

where Y_t denotes the first-differenced Brent return at date t, p is the lag order, α_0 is an intercept, and e_t is the error term. The unrestricted model augments this specification with lagged values of the entropy indicator:

$$Y_t = \alpha_0 + \sum_{j=1}^{p} \alpha_j Y_{t-j} + \sum_{j=1}^{p} \beta_j X_{t-j} + u_t, \tag{5}$$

where X_{t-j} denotes the first-differenced entropy indicator at lag j, and $\beta_1, \ldots, \beta_p$ are the coefficients on the lagged indicator values. The null hypothesis is that the indicator carries no incremental predictive content:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0. \tag{6}$$

Under this null, the lagged entropy values are jointly irrelevant for predicting future returns once the returns' own history is accounted for. Rejection of H_0 constitutes evidence that the entropy indicator Granger-causes Brent returns, in the specific sense that it carries information about future returns that is not already embedded in the returns' own autoregressive structure.

The test statistic is a standard F-test comparing the residual sum of squares of the restricted and unrestricted models:

$$F = \frac{(\mathrm{RSS}_r - \mathrm{RSS}_{ur})/p}{\mathrm{RSS}_{ur}/(T - 2p - 1)}, \tag{7}$$

where RSS_r and RSS_{ur} are the residual sums of squares of the restricted and unrestricted models respectively, p is the number of lags (equal in both the Y and X lag polynomials), and $T = N - p$ is the effective sample size after accounting for the lag truncation. Under H_0, this statistic follows an $F(p, T - 2p - 1)$ distribution (Granger, 1969; Hamilton, 1994, Chapter 11).

The lag order p is selected by the Akaike Information Criterion (AIC), which is the standard choice when the primary objective is predictive accuracy rather than model parsimony (Akaike, 1974). The AIC balances goodness of fit against model complexity by penalising each additional parameter:

$$\mathrm{AIC}(p) = \ln\left(\frac{\mathrm{RSS}_{ur}}{T}\right) + \frac{2k}{T}, \tag{8}$$

where $k = 2p + 1$ is the total number of estimated parameters in the unrestricted model (the intercept plus p own lags plus p indicator lags). The criterion is evaluated for every candidate lag order from $p = 1$ to a maximum of $p_{\max} = 12$ trading days (approximately two and a half trading weeks), and the lag order that minimises the AIC is selected for the F-test. The choice of $p_{\max} = 12$ ensures that the test can detect lead-lag relationships extending over a meaningful economic horizon without consuming an excessive number of degrees of freedom in a daily-frequency sample.

It is important to note what Granger causality does and does not establish. The test provides evidence of incremental predictive content in a forecasting sense: lagged entropy values carry information about future returns that is not already captured by the returns' own lags. It does not establish structural or economic causation (Granger, 1969). The entropy indicator may lead returns because it captures the early stages of a distributional regime shift that subsequently manifests in realised price movements, or because both series respond to a common underlying driver with different speeds. The test is agnostic about the mechanism; it speaks only to the statistical predictability that an early warning system requires.

1.5 Internal Coherence of the Entropy Indicators

Before interpreting the lead-lag results, it is informative to examine whether the three entropy components, full, left-tail, and right-tail, move together or exhibit distinct dynamics. This is assessed by computing the pairwise CCF among all three indicators. If the components share a common factor, their pairwise CCFs should peak near zero lag with high correlation. If they respond to different types of shocks with different timing, the pairwise CCFs may reveal non-trivial lead-lag structure among the entropy measures themselves.

The coherence test applied here follows the internal-consistency diagnostic used in composite-indicator construction (Bujosa, García-Ferrer, and de Juan, 2013). In that framework, a well-constructed composite leading indicator should have components that are internally coherent, meaning they move together contemporaneously ($h^* \approx 0$) rather than leading or lagging one another. The expectation for the entropy indicators is similar but not identical: the full-distribution entropy is by construction a function of both tails, so it should be contemporaneously correlated with each tail component. The left-tail and right-tail entropies, however, respond to different types of shocks, downside and upside respectively, and need not be correlated with each other at all.

1.6 Results

The early warning battery is applied to the three rolling-window entropy indicators (full, left-tail, and right-tail KL divergence against the Normal baseline) computed at $h = 5$ over the out-of-sample period from June 2012 to April 2026, comprising 2,167 daily observations after first-differencing.

1.6.1 Cross-Correlation Function Results

The CCF analysis reveals that all three entropy indicators lead Brent returns, with the peak correlation occurring at strictly positive lags in every case.

The full-distribution entropy exhibits its peak cross-correlation at $h^* = +15$ trading days with $r = 0.311$, well above the Bartlett 95 percent confidence band of ± 0.042. This result indicates that a spike in the overall divergence of the conditional density from the Normal reference today is most strongly associated with return movements approximately three trading weeks ahead. The right-tail entropy peaks at $h^* = +7$ with $r = 0.285$, indicating that upside vulnerability signals propagate to returns more quickly, within roughly one and a half trading weeks. The left-tail entropy peaks at $h^* = +16$ with $r = -0.149$; the negative sign reflects the inverse relationship between left-tail fattening (which represents an accumulation of downside risk) and subsequent returns (which tend to decline when downside risk materialises), and the longer lead time of 16 trading days is consistent with the observation that downside risk builds gradually as geopolitical tensions escalate or demand conditions deteriorate before the price impact is fully absorbed.

The fact that all three indicators exhibit $h^* > 0$ is the central result of the CCF analysis: the entropy measures are leading indicators, not coincident or lagging diagnostics. The varying lead times across the three components carry economic content. The right-tail entropy, which captures speculative upside pressure, translates to returns within about a week, consistent with the speed at which supply-disruption premia are priced into the futures curve. The left-tail and full-distribution entropies, which capture broader distributional stress including the accumulation of downside vulnerability, require two to three weeks to manifest in realised returns, consistent with the slower propagation of demand-side shocks and the gradual repricing of risk premia during periods of elevated macroeconomic uncertainty.

1.6.2 Granger Causality Results

The Granger causality tests confirm and strengthen the CCF findings. The AIC selects a lag order of $p = 12$ for all three indicators, indicating that the predictive relationship extends over approximately two and a half trading weeks.

Table 1: Granger Causality Test Results: Entropy Indicators → Brent Returns

Indicator	F-statistic	p-value	Selected Lag (p)	Significance
Full Entropy	41.960	< 0.001	12	***
Left Entropy	3.940	< 0.001	12	***
Right Entropy	47.593	< 0.001	12	***

*** significant at the 1% level. Lag order selected by AIC with $p_{\max} = 12$.
Sample: 2,167 daily observations (June 2012 – April 2026), first-differenced.
Source: Granger (1969); lag selection via Akaike (1974).

All three entropy indicators Granger-cause Brent returns at the 1 percent significance level. The null hypothesis that the lagged entropy values carry no incremental predictive content beyond the returns' own autoregressive structure is rejected decisively in every case, with p-values below 10^{-5}.

The F-statistics reveal an instructive ordering of predictive power. The right-tail entropy exhibits the strongest Granger-causal relationship ($F = 47.593$), followed by the full-distribution entropy ($F = 41.960$) and the left-tail entropy ($F = 3.940$). The dominance of the right-tail entropy is economically interpretable: in crude oil markets, the upside tail of the return distribution is disproportionately driven by supply-side shocks, geopolitical escalation, and speculative positioning, all of which tend to be abrupt and to generate sharp, directional price movements. The conditional density detects the onset of these episodes through the fattening of the right tail, and the Granger test confirms that this signal carries information about future returns that the returns' own history does not.

The left-tail entropy, while still statistically significant, exhibits a considerably lower F-statistic. This is consistent with the nature of downside risk in oil markets: demand-side contractions tend to unfold more gradually than supply disruptions, and the return series itself, through its own autoregressive structure, already absorbs much of the predictable component of these slow-moving declines. The incremental information contributed by the left-tail entropy is therefore smaller in magnitude, although still statistically significant.

1.6.3 Internal Coherence

The pairwise CCF analysis among the three entropy indicators reveals the expected pattern of partial coherence.

Table 2: Internal Coherence: Pairwise CCF Among Entropy Indicators

Series X	Series Y	h^*	r at h^*	Significant
Full Entropy	Left Entropy	0	$+0.748$	Yes
Full Entropy	Right Entropy	0	$+0.631$	Yes
Left Entropy	Right Entropy	0	-0.036	No

Bartlett 95% confidence band: $\pm 1.96/\sqrt{N}$.

The full-distribution entropy is strongly and contemporaneously correlated with both the left-tail component ($r = 0.748$, $h^* = 0$) and the right-tail component ($r = 0.631$, $h^* = 0$). This is expected by construction, since the full entropy integrates over the entire support of the density, including both tails. The left-tail and right-tail entropies, however, are essentially uncorrelated ($r = -0.036$, not significant), confirming that they respond to distinct economic phenomena. A supply-disruption shock that fattens the right tail of the return distribution does not simultaneously fatten the left tail, and a demand-contraction episode that elevates left-tail entropy does not generate upside vulnerability. The two tail components carry independent information, which is why the early warning system benefits from monitoring both rather than relying on the full-distribution aggregate alone.

1.7 Synthesis and Implications

The combined evidence from the Cross-Correlation Function and the Granger causality test supports the conclusion that the entropy measures derived from the Oil-at-Risk framework function as effective early warning indicators of oil price movements. The key findings are as follows.

First, all three entropy indicators, full-distribution, left-tail, and right-tail, lead Brent returns, with optimal lead times ranging from 7 to 16 trading days. The entropy measures are not merely contemporaneous diagnostics of distributional stress; they anticipate future return movements with a lead time that is operationally meaningful for risk management.

Second, the Granger causality tests confirm that the predictive content of the entropy indicators is incremental: it cannot be replicated by the returns' own autoregressive structure. The entropy measures capture information about the evolving shape of the conditional distribution that is not yet reflected in realised price movements.

Third, the right-tail entropy provides the strongest and fastest signal, consistent with the abrupt, event-driven nature of supply-side shocks in crude oil markets. The left-tail entropy provides a weaker but still statistically significant signal, consistent with the slower propagation of demand-side vulnerabilities. Monitoring both tails separately, rather than relying on the full-distribution aggregate, provides a richer early warning profile that can distinguish between upside and downside stress.

Fourth, the internal coherence analysis confirms that the left-tail and right-tail entropies respond to distinct economic phenomena. They are essentially uncorrelated at all lags, indicating that the two components carry independent information about different types of oil market risk.

From a practical perspective, the early warning system implies the following operational protocol. A risk manager monitoring the Oil-at-Risk dashboard observes the daily rolling entropy series decomposed by tail. A significant spike in right-tail entropy signals that the conditional distribution is developing an unusually fat right tail, typically driven by geopolitical escalation or supply tightening, and that this stress is likely to manifest in realised price movements within approximately 7 trading days. A significant spike in left-tail entropy signals the accumulation of downside vulnerability, typically associated with weakening demand conditions or an appreciating dollar, with an anticipation window of approximately 16 trading days. The full-distribution entropy, which aggregates both tails, provides the broadest signal with a 15-day lead time.

These lead times are sufficiently long to permit meaningful risk management responses: adjusting hedge ratios, recalibrating margin requirements, or revising scenario assumptions for stress testing. The entropy-based early warning system therefore closes the operational loop of the Oil-at-Risk framework, transforming the distributional analysis from a retrospective diagnostic into a forward-looking surveillance tool.

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